

Exploring and Improving Cross- and Multi-Domain Personalized Question Answering via a Dual Retrieval Augmentation Approach

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Abstract

Personalized long-form question answering aims to tailor responses based on a user’s historical data, which often spans diverse information domains. While prior work typically assumes user profiles are domain-aligned with the users’ questions, real-world profiles are often diverse, containing information across multiple domains. In this work, we use the LaMP-QA benchmark, where user profiles span diverse Stack Exchange domains (Arts & Entertainment, Lifestyle & Personal Development, and Society & Culture), to systematically investigate how domain composition of the user profile affects personalization performance. We examine three profile composition settings: *Cross-Domain*, where profiles contain only out-of-domain information; *Multi-Domain*, where profiles span multiple domains including the question’s domain; and *In-Domain*, where profiles match the question domain. Our analysis reveals that state-of-the-art personalization methods struggle in *Cross-Domain* settings, often failing to outperform non-personalized baselines. To address this challenge, we introduce Dual Retrieval Augmentation (D-RA), a framework that jointly retrieves from both the user profile and a public corpus. We apply D-RA to existing personalization approaches and demonstrate that it consistently outperforms state-of-the-art methods across all three profile settings, validating its robustness for personalized question answering.

CCS Concepts

• Information systems → Personalization.

Keywords

Personalization, Retrieval Augmented Generation, Large Language Models

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1 Introduction

Personalization is a fundamental component of modern information systems, playing a key role in search [19], recommendation [10], and text generation [15, 18]. By tailoring outputs to a user’s intent, background, and preferences, personalized systems can improve user satisfaction and trust. With the rise of Large Language Models (LLMs) that interact with users through natural language, *personalized text generation* has become an increasingly important area of study.

Recent efforts have introduced benchmarks for personalized generation, progressing from content-focused tasks such as personalized tweet paraphrasing in LaMP [15] or personalized email generation in LongLaMP [7] to the more recent LaMP-QA benchmark [16] that focuses on personalized question answering (QA). Additionally, a variety of frameworks have been proposed to personalize LLMs, ranging from Retrieval Augmented Generation (RAG) using user information [15] to training with personalized human feedback [13], optimizing LLMs with user-specific supervision [5], creating personalized prompts [8], and parameter-efficient fine-tuning solutions per user [17].

A common assumption in personalized question answering is that a user’s historical interactions are topically aligned with their current query. Under this assumption, retrieving relevant items from the user profile provides useful context for generating a personalized response. In practice, however, users often have diverse interests that span multiple topics and domains. A user who asks a question about music production may have a history that includes questions about cooking, finance, and philosophy. This mismatch between the topical scope of the profile and the domain of the current question raises an important but underexplored research question: *How does the domain composition of a user’s profile affect personalization performance?*

Our contributions can be summarized as follows: (1) We present the first systematic study of how user profile domain composition affects personalized long-form QA. (2) We show that existing personalization approaches degrade substantially in certain profile composition settings, often underperforming non-personalized baselines. (3) We propose Dual Retrieval Augmentation (D-RA), a general framework that jointly retrieves from the user profile and a public corpus containing target-domain content. We demonstrate the generality of D-RA by applying it to both RAG-based personalization [15] and planning-based approaches [16], achieving statistically significant improvements over state-of-the-art baselines across all three profile composition settings.

To improve reproducibility and foster research in this area, we open-source our implementation and publicly release our experiment results at [this repository](#).

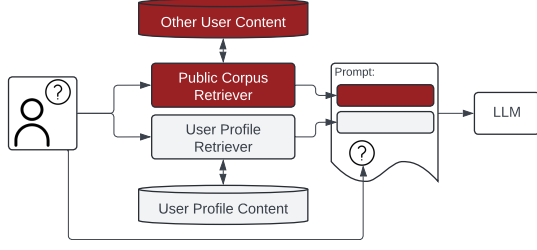


Figure 1: General retrieval architecture comparison. Standard RAG (Gray) retrieves only from the user profile. D-RA framework (Maroon + gray) performs personalized retrieval and public corpus retrieval.

2 Methodology

We first formalize the problem of personalized QA across different domain composition settings (§2.1), then introduce our core idea, Dual Retrieval-Augmentation (§2.2), and show how it can be applied to two existing personalization approaches (§2.3–§2.4).

2.1 Problem Formulation

We consider a scenario where a user u poses a question x_u . Following prior work that incorporates long-term user history as a user profile [6, 15], we assume the user profile $P_u = \{p_i\}_{i=1}^{n_u}$ consists of n_u previously asked questions along with the detailed narrative of the question written by the user. Each profile entry p_i is associated with a domain label $\delta(p_i) \in \mathcal{D}$, where $\mathcal{D}$ represents the set of all domains. Similarly, the user’s current question x_u belongs to a target domain $\delta(x_u)$. We define the domain-specific subset of a user profile as $P_u^{(t)} = \{p \in P_u : \delta(p) = t\}$. Lastly, we define C to be a public corpus that contains posts from other users in the target domain ($\delta(x_u)$). We investigate three user profile settings based on the relationship between the profile domains and the question domain:

Cross-Domain. The user profile contains entries only from domains other than the question domain: $P_u \cap P_u^{(\delta(x_u))} = \emptyset$.

Multi-Domain. The user profile contains all available user entries, without domain filtering.

In-Domain. The user profile contains entries exclusively from the question domain: $P_u = P_u^{(\delta(x_u))}$.

The goal of personalized long-form question answering is to generate a personalized response $y_u = f(x_u, P_u, C)$.

2.2 Dual Retrieval-Augmentation

Recent work has shown that RAG provides a strong framework for personalizing LLMs [7, 15, 16]. These methods retrieve relevant entries from the user profile P_u and condition the generator on them. However, our analysis reveals that this approach suffers substantial performance degradation in cross-domain and some multi-domain settings, often underperforming non-personalized baselines (§3.4). We attribute this to a *domain gap*: when the user profile lacks content from the target question domain, retrieved out-of-domain entries provide insufficient and potentially misleading context.

To address this, we propose Dual Retrieval-Augmentation (D-RA), which supplements the user profile retrieval with a second

retrieval from a public corpus C containing content in the target domain (Figure 1). The two retrieval sources play complementary roles: the user profile captures individual preferences and communication style, while the public corpus supplies domain-specific language and concepts that bridge the domain gap. Together, they allow the generator to produce responses that are both personalized and grounded in the target domain.

D-RA is a general principle, not a specific model. In the remainder of this section, we apply it to two existing personalization approaches.

2.3 Dual Retrieval-Augmented Generation (D-RAG)

We denote a language model M and a retrieval function $R(q, S, k)$ that returns k entries from corpus S given query q . Standard RAG-based personalization generates a response as $\hat{y}_u = M(x_u, R(x_u, P_u, k))$. D-RAG extends this by additionally conditioning on public retrieval:

$$\hat{y}_u = M(x_u, R(x_u, P_u, k), R(x_u, C, k)) \quad (1)$$

2.4 Planned Dual Retrieval-Augmented Generation (Plan-D-RAG)

Recent work has shown that planning-based generation, in which a model first predicts the key aspects a response should address, improves long-form personalized QA [16]. We apply D-RA to this paradigm, yielding Plan-D-RAG.

Plan-D-RAG uses a fine-tuned planner M_{plan} that produces a structured aspect list before the generator writes the final answer. Critically, both the planner and the generator are conditioned on dual retrieval context, enabling them to synthesize personalized preferences with target-domain content. Generation proceeds in two stages:

$$\text{plan} = M_{\text{plan}}(x_u, R(x_u, P_u, k), R(x_u, C, k)) \quad (2)$$

$$\hat{y}_u = M(x_u, R(x_u, P_u, k), R(x_u, C, k), \text{plan}) \quad (3)$$

The planner is trained following the framework introduced in LaMP-QA [16], we refer readers there for full details on planner training and aspect prediction.

3 Experiments

3.1 Datasets and Evaluation

We conduct our experiments on the LaMP-QA benchmark [16], a recent personalized long-form QA benchmark consisting of user profiles from the Stack Exchange platform. Each user profile contains questions from three broad Stack Exchange categories: Arts and Entertainment (A&E), Lifestyle and Personal Development (L&P), and Society and Culture (S&C). This multi-domain structure makes LaMP-QA particularly well suited for our study, as it enables systematic investigation of in-, cross-, and multi-domain personalization settings. For evaluating personalization performance, we follow the aspect-based evaluation framework from LaMP-QA [9, 14, 16], which extracts personalization aspects from question narratives and grades generated answers on each aspect using an LLM evaluator (Qwen 2.5 32B [12]). Note that the narratives and extracted aspects are hidden from generator models during inference to ensure unbiased evaluation. Specifically, each aspect is scored on a

Table 1: Test set performance across different domain settings. A&E, L&P, and S&C denote the target domain of the question. Bold indicates best results. [†] denotes statistical significance against all other baselines, and * denotes statistical significance against RAG counterpart method (paired t-test, Bonferroni Correction applied, $p < 0.05$).

Model	Method	Cross Domain				Multi Domain				In Domain			
		A&E	L&P	S&C	Avg	A&E	L&P	S&C	Avg	A&E	L&P	S&C	Avg
Qwen 2.5 (3B)	No-Personalization	0.281	0.4364	0.4518	0.3897	0.281	0.4364	0.4518	0.3897	0.281	0.4364	0.4518	0.3897
	Public-Only	0.2275	0.3759	0.3952	0.3328	0.2275	0.3759	0.3952	0.3328	0.2275	0.3759	0.3952	0.3328
	RAG	0.1907	0.308	0.3092	0.2693	0.2757	0.3708	0.4107	0.3524	0.2684	0.3718	0.4106	0.3503
	Plan-RAG	0.2844	0.4123	0.4338	0.3768	0.3318	0.4465	0.4679	0.4154	0.3322	0.4386	0.4812	0.4173
	D-RAG	0.2659*	0.4002*	0.4318*	0.3659*	0.2858	0.4231*	0.4458*	0.3849*	0.2823	0.4214*	0.4503*	0.3847*
	Plan-D-RAG	0.3149[†]*	0.4295*	0.4595*	0.4013[†]*	0.3329	0.4478	0.476	0.4189	0.3362	0.4406	0.4719	0.4162
Qwen 2.5 (7B)	No-Personalization	0.3196	0.4662	0.4794	0.4217	0.3196	0.4662	0.4794	0.4217	0.3196	0.4662	0.4794	0.4217
	Public-Only	0.3684	0.489	0.5193	0.4589	0.3684	0.489	0.5193	0.4589	0.3684	0.489	0.5193	0.4589
	RAG	0.2615	0.3996	0.4346	0.3652	0.3424	0.4486	0.5011	0.4307	0.34	0.4451	0.5027	0.4293
	Plan-RAG	0.3264	0.4642	0.4958	0.4288	0.3653	0.488	0.5305	0.4612	0.3648	0.4827	0.535	0.4608
	D-RAG	0.3319*	0.4803 [†] *	0.4958*	0.436*	0.3558	0.4855*	0.4963	0.4458*	0.3524	0.4837*	0.5016	0.4459*
	Plan-D-RAG	0.3607*	0.491[†]*	0.5243[†]*	0.4587*	0.3745	0.5008*	0.5398	0.4717[†]*	0.3801*	0.5068[†]*	0.5361	0.4743[†]*
Qwen 2.5 (14B)	No-Personalization	0.3325	0.4882	0.5058	0.4421	0.3325	0.4882	0.5058	0.4421	0.3325	0.4882	0.5058	0.4421
	Public-Only	0.3399	0.4883	0.5047	0.4443	0.3399	0.4883	0.5047	0.4443	0.3399	0.4883	0.5047	0.4443
	RAG	0.3043	0.4311	0.4526	0.396	0.3902	0.4866	0.5379	0.4715	0.3897	0.4805	0.5358	0.4687
	Plan-RAG	0.3513	0.4614	0.506	0.4395	0.4019	0.4934	0.5639	0.4864	0.4054	0.4888	0.5612	0.4851
	D-RAG	0.3371*	0.4848*	0.5184*	0.4467*	0.3665	0.4915	0.5309	0.4629	0.3671	0.4871	0.5308	0.4617
	Plan-D-RAG	0.3978[†]*	0.5038[†]*	0.5621[†]*	0.4879[†]*	0.4092	0.5113[†]*	0.5705	0.497[†]*	0.407	0.506[†]*	0.5733*	0.4954[†]*
GPT-4o-mini	No-Personalization	0.3797	0.5159	0.5252	0.4736	0.3797	0.5159	0.5252	0.4736	0.3797	0.5159	0.5252	0.4736
	Public-Only	0.4272	0.528	0.5761	0.5104	0.4272	0.528	0.5761	0.5104	0.4272	0.528	0.5761	0.5104
	RAG	0.3089	0.4401	0.4479	0.3989	0.3954	0.4778	0.5308	0.468	0.4042	0.4818	0.5262	0.4707
	Plan-RAG	0.3949	0.5275	0.5546	0.4923	0.453	0.5484	0.6091	0.5368	0.4604	0.5475	0.6095	0.5391
	D-RAG	0.396*	0.5469 [†] *	0.5771*	0.5066*	0.4223*	0.5557*	0.6067*	0.5282*	0.4239*	0.5561*	0.6069*	0.5289*
	Plan-D-RAG	0.4494[†]*	0.5523[†]*	0.6093[†]*	0.537[†]*	0.4559[†]	0.5645[†]*	0.6226[†]*	0.5476[†]*	0.46	0.5573*	0.6211[†]*	0.5461[†]*

scale from 0 to 2 and then normalized by dividing by 2. The final evaluation score for a generated response is the average of the normalized aspect scores. We refer readers to LaMP-QA [16] for additional details on the prompts used in evaluation.

Public Corpus Construction. Our framework requires a public corpus C that excludes the current user’s content and always contains target-domain information. We construct C by pooling all target domain profile entries from other users in the dataset split. However, LaMP-QA contains duplicate user profiles where the same real-world user appears with different user IDs. To prevent leakage, we implement checksum-based filtering: for each user u , we compute checksums for all posts and exclude any profile sharing matching checksums with u during retrieval, this ensures any retrieved content from C is not from the current user.

3.2 Experimental Setup

We use Qwen2.5 (3B, 7B, 14B) [12] and GPT-4o-mini [11] as the generator LLM M . For the planner model M_{plan} , we fine-tune Qwen2.5-7B using LoRA [3] with a sequence-to-sequence cross-entropy objective. All models use a maximum context length of 8192 tokens and generate with temperature 0 [2]. For retrieval, we use Contriever [4] fine-tuned on MS MARCO [1] to retrieve $k = 10$ entries from both the user profile and public corpus.

3.3 Baselines

We compare against four baselines that represent a progression from no personalization to planned personalization. All baselines except for Public-Only are drawn from the publicly available LaMP-QA Benchmark [16].

No-Personalization. The question x_u is provided directly to the LLM without additional context: $\hat{y}_u = M(x_u)$.

Public-Only. The top k retrieved posts from the public corpus C and provided as context: $\hat{y}_u = M(x_u, R(x_u, C, k))$. This represents standard RAG without personalization (green segments in Figure 1).

RAG. The top- k posts are retrieved from the user profile P_u : $\hat{y}_u = M(x_u, R(x_u, P_u, k))$. This is the single-source counterpart to D-RAG.

Plan-RAG. A fine-tuned planner M_{plan} generates aspects based on the question and retrieved profile content, which then guide response generation: $\hat{y}_u = M(x_u, R(x_u, P_u, k), M_{\text{plan}}(x_u, R(x_u, P_u, k)))$. This is the single-source counterpart to Plan-D-RAG and is the current state-of-the-art method for personalization.

3.4 Empirical Findings

Table 1 presents results across the three profile settings, revealing several consistent patterns. First, Plan-D-RAG achieves the best average performance in the majority of experimental configurations, demonstrating the effectiveness of combining dual retrieval with planning. Second, methods incorporating user profile retrieval substantially outperform public-only retrieval in nearly all cases,

highlighting the critical importance of user-specific context for personalization. Third, D-RAG and Plan-D-RAG methods outperform their single retrieval baselines in the majority of configurations and consistently improves with model capacity. Larger models appear better equipped to synthesize information from multiple retrieval sources without distraction from potentially irrelevant content.

Cross-Domain Results. The Cross-Domain setting represents a challenging scenario, where user profiles contain no information from the target question domain. RAG and Plan-RAG perform poorly, consistently falling below the No-Personalization baseline across all model sizes, indicating that retrieving from misaligned profile content actively harms generation quality. Our analysis reveals that dual augmentation with public data in the target domain is essential across all domains: when retrieved profile content is misaligned with the user’s question, it guides the generator LLM to form incorrect conclusions and spurious connections. This is particularly prominent in sub-categories, such as hobbies and work, where users often reference specific content (e.g., particular TV series or video games) from previous discussions. Without target domain context, the generator cannot accurately identify which specific interests the user is referring to. We verify the importance of target domain content by measuring retrieval precision for each user, treating posts from the target domain as relevant and others as non-relevant, then averaging across users. The average P@10 was 0.7834 (A&E: 0.8661, L&P: 0.6581, S&C: 0.8261), indicating that more than half of retrieved posts from user profiles belong to the target domain. In the Cross-Domain setting, we deliberately exclude these relevant posts, explaining why Cross-Domain personalization presents such a significant challenge.

Our proposed D-RAG method substantially improves over standard RAG by incorporating public information containing the target domain, recovering much of the lost performance. Notably, Public-Only often outperforms RAG in cross-domain settings, suggesting that when user profiles are domain-misaligned, personalization attempts can be counterproductive. D-RAG effectively combines the benefits of both approaches, matching or exceeding Public-Only performance while retaining the ability to leverage user-specific signals. Plan-D-RAG further extends these gains, consistently matching or exceeding all baselines, with larger models showing the most substantial improvements.

Multi-Domain and In-Domain Results. The Multi-Domain setting represents the most realistic scenario for real-world personalization systems, where user profiles span multiple domains including the target. Interestingly, performance here is comparable to the In-Domain setting, and in some cases slightly exceeds it, suggesting that diverse user profiles are not inherently detrimental; rather, additional context from related domains may provide valuable supplementary information. The relative benefit of dual retrieval remains substantial across all model sizes, with Plan-D-RAG consistently outperforming both standard RAG and Plan-RAG. In the In-Domain setting, where user profiles contain exclusively information from the target domain, standard RAG performs substantially better, validating that domain alignment is crucial for effective personalization. The contrast between RAG performance in Cross-Domain versus In-Domain settings underscores a central challenge this work addresses. Plan-D-RAG continues to outperform all baselines even in

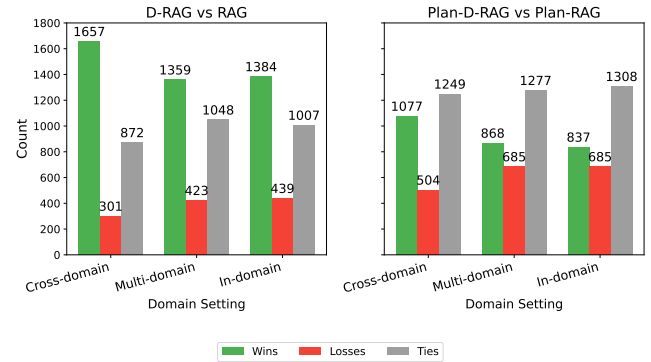


Figure 2: Win/Loss/Tie Counts of dual augmentation against the baseline model (GPT-4o-mini).

this favorable scenario, indicating that supplementing user profiles with public knowledge provides complementary information even when profiles are already domain-aligned. The performance gap between Plan-RAG and Plan-D-RAG is notably smaller in this setting compared to Cross-Domain, suggesting that the planning mechanism with D-RAG framework provides greater marginal benefit when compensating for missing domain knowledge.

Instance-Level Analysis. While aggregate metrics demonstrate the effectiveness of the D-RA framework, Figure 2 reveals important findings at the instance level. Comparing D-RAG against RAG using GPT-4o-mini, D-RAG wins on substantially more instances than it loses across all settings, with the advantage most pronounced in the Cross-Domain setting. This win rate narrows in Multi-Domain and In-Domain settings, suggesting that dual retrieval provides the greatest marginal benefit when user profiles lack target domain content. However, a non-trivial number of instances exhibit performance degradation with dual retrieval when comparing planning-based personalization methods. Our analysis reveals that the planner generates aspects related to public knowledge at the expense of more valuable user-specific information, producing plans that diverge from the optimal aspects. This suggests that while the planning mechanism is beneficial overall, it may be more sensitive to the balance between personalization and target domain knowledge. These findings motivate several promising directions for future work, including adaptive retrieval strategies that dynamically weight or select between sources based on query characteristics, as well as separate planning trajectories for public and personal corpora.

4 Conclusion

This paper systematically analyzed personalized question answering using the LaMP-QA benchmark across three domain composition settings: Cross-Domain, Multi-Domain, and In-Domain. We demonstrated that state-of-the-art personalization methods struggle when user profiles lack target domain content, often performing worse than non-personalized baselines in Cross-Domain scenarios. To address this challenge, we introduced Dual Retrieval-Augmentation (D-RA), a framework that jointly retrieves from both

user profiles and a public corpus to influence the model distribution towards the target domain. Experiment results demonstrate that methods using D-RA framework consistently improves performance across all domain composition settings, with particularly substantial gains in challenging Cross-Domain scenarios. We further showed that combining dual retrieval with planning-based generation (Plan-D-RAG) achieves the best overall performance. Our findings highlight the importance of considering domain composition in personalization and demonstrate that leveraging multiple information sources provides a robust solution for personalized question answering with diverse user profiles.

5 Limitations

Our study has several limitations that suggest directions for future work. Mainly, all of our experiments use models from the Qwen 2.5 family for both generation and evaluation. While this provides a controlled experimental setting, further research is needed to verify whether the observed failure of personalization in cross-domain and multi-domain settings, as well as the effectiveness of D-RA, generalize across other model families and scales.

Second, while the cross-domain setting may appear extreme or artificial, it mirrors a realistic cold-start scenario in which a user has no prior experience in the target domain, or too little relevant history for meaningful retrieval. Investigating how D-RA performs under varying degrees of domain overlap ranging from full cold-start to sparse in-domain coverage, would provide a more complete picture of its practical applicability.

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